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Solidification of Al-Bi-Sn immiscible alloy under microgravity conditions of space

Wang Li [a,b], Hongxiang Jiang [a,*], Lili Zhang [a], Shixin Li [a,b], Jie He [a], Jiuzhou Zhao [a,b,**], Fei Ai [c]

[a] Institute of Metal Research, Chinese Academy of Sciences, Shenyang 110016, PR China
[b] School of Materials Science and Engineering, University of Science and Technology of China, Shenyang 110016, PR China
[c] State Key Laboratory of High Performance Ceramics and Superfine Microstructure, Shanghai Institute of Ceramics, Chinese Academy of Sciences, Shanghai 200050, PR China

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ABSTRACT

Al-Bi-Sn alloy was solidified directionally in space. The sample shows a well-dispersed microstructure. A model was developed to describe the microstructure formation during solidification. Calculations demonstrate that the Stokes settlement of the Minority Phase Droplets (MPDs) and the convection of the melt are against the establishment of the steady state of solidification and macro-segregation occurs soon after the beginning of the solidification on ground. The diminished melt convection and the Stokes motions of the MPDs under microgravity conditions promote the establishment of the steady state of solidification and the formation of a well-dispersed microstructure.

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Immiscible alloys are characterized by the appearance of a miscibility gap in the liquid state. Many of them have great potentials to be used as important structural and functional materials [1,2]. However, these alloys transform into two liquids during a cooling through the miscibility gap and generally show a phase segregated microstructure when solidified on ground [3,4]. Since last century, solidification experiments were carried out in space to get rid of gravity effects and obtain immiscible alloys with a dispersed microstructure. Unfortunately, most of the experiments failed in doing so. For examples, Ahlborn et al. [5], Potard [6] and Gelles et al. [7] solidified Al-In alloys in space and found that the sample showed a microstructure with serious phase segregation. Fredriksson et al. [8] solidified Zn-Bi alloys under microgravity conditions and found that all the samples showed a microstructure with macro-segregation. The extent of the macro-segregation increased with the increase of the Bi content. Luo et al. [9,10] did directional solidification experiment with Al-7 wt%Bi alloy on Shenzhou III. They found that although the inner part of the sample showed a dispersed microstructure [9], a Bi-rich layer appeared on part of the surface of the sample and also the last solidified region was rich in Bi [10]. Stimulated by these un-expected results, scientists carried out a lot of experiments to investigate the solidification of immiscible alloys by using different methods/techniques, i.e., rapid solidification [11], continuous/directional solidification [12] and solidification under the effect of the external fields [13–16], etc. Meanwhile, models were built to describe the microstructure evolution [11,17–21]. The results demonstrate that the microstructure formation during the liquid-liquid phase transformation is a result of the concurrent actions of the nucleation, growth, Ostwald ripening and motions of the Minority Phase Droplets (MPDs). It is very complex. Especially the convective flow mixes up the effects of different factors and makes it extremely difficult to uncover the details of the microstructure evolution.

Recently we carried out solidification experiments with Al-Bi-Sn alloy in space during the flight of the Tiangong II spacecraft of China. Through properly designing the alloy composition and experimental scheme based on the extensive pre-research, the alloy solidified at a cooling rate as low as ~0.026 °C/s shows a well dispersed microstructure. The microstructure formation was calculated and the microgravity effects were discussed.

The solidification conditions provided by the Material Experimental Furnace assembled on the Tiangong II spacecraft of China is: working temperature ≤ 700 °C, solidification velocity ≤ 28 μm/s. Al-Bi was chosen as the basic experimental alloy system considering its relatively low binodal line temperature. Based on our pre-research [22], Sn was used as the ternary element to reduce the interfacial energy (σ_{L_1/L_2}) between the two liquid phases and, thus, enhance the nucleation rate and weaken the Marangoni migration of the MPDs during the liquid-liquid decomposition. The composition of the alloy was chosen as Al-3.6 wt% Bi-1.0 wt%Sn, whose binodal line temperature is 658.9 °C (see Fig. S3 in Appendix A). Also, the temperature for the heating region B of the

* Corresponding author.
** Correspondence to: J. Zhao, Institute of Metal Research, Chinese Academy of Sciences, Shenyang 110016, PR China.
E-mail addresses: hxjiang@imr.ac.cn (H. Jiang), zzhao@imr.ac.cn (J. Zhao).

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furnace is set 15 °C lower than that for the heating region A (see Fig. S1 in Appendix A) to reduce the temperature gradient in front of the solidification interface and, thus, the Marangoni migration of the MPDs.

Solidification experiments were carried out both on ground and in space under the atmosphere of 1 atm. The experimental procedures were as follows: Firstly, the alloy was melted, heated to 685 °C and held temperature for 150 min to form a single-phase homogeneous liquid. The ampoule was then withdrawn out of the furnace at a rate of 28 μm/s for solidification. For the ground sample, the ampoule was withdrawn against the gravity direction. The detailed experimental procedures were given in Appendix A.

After solidification, X-ray tomography (XRT) was first performed to analyze the spatial distribution of the Minority Phase Particles (MPPs) in the samples. The samples were then sectioned both longitudinally and transversely for the preparation of the metallographic specimens. The microstructure was analyzed by using the optical microscope (OM) and scanning electron microscope (SEM) equipped with an energy dispersive X-ray spectrometry (EDS). The SISC IAS V 8.0 software

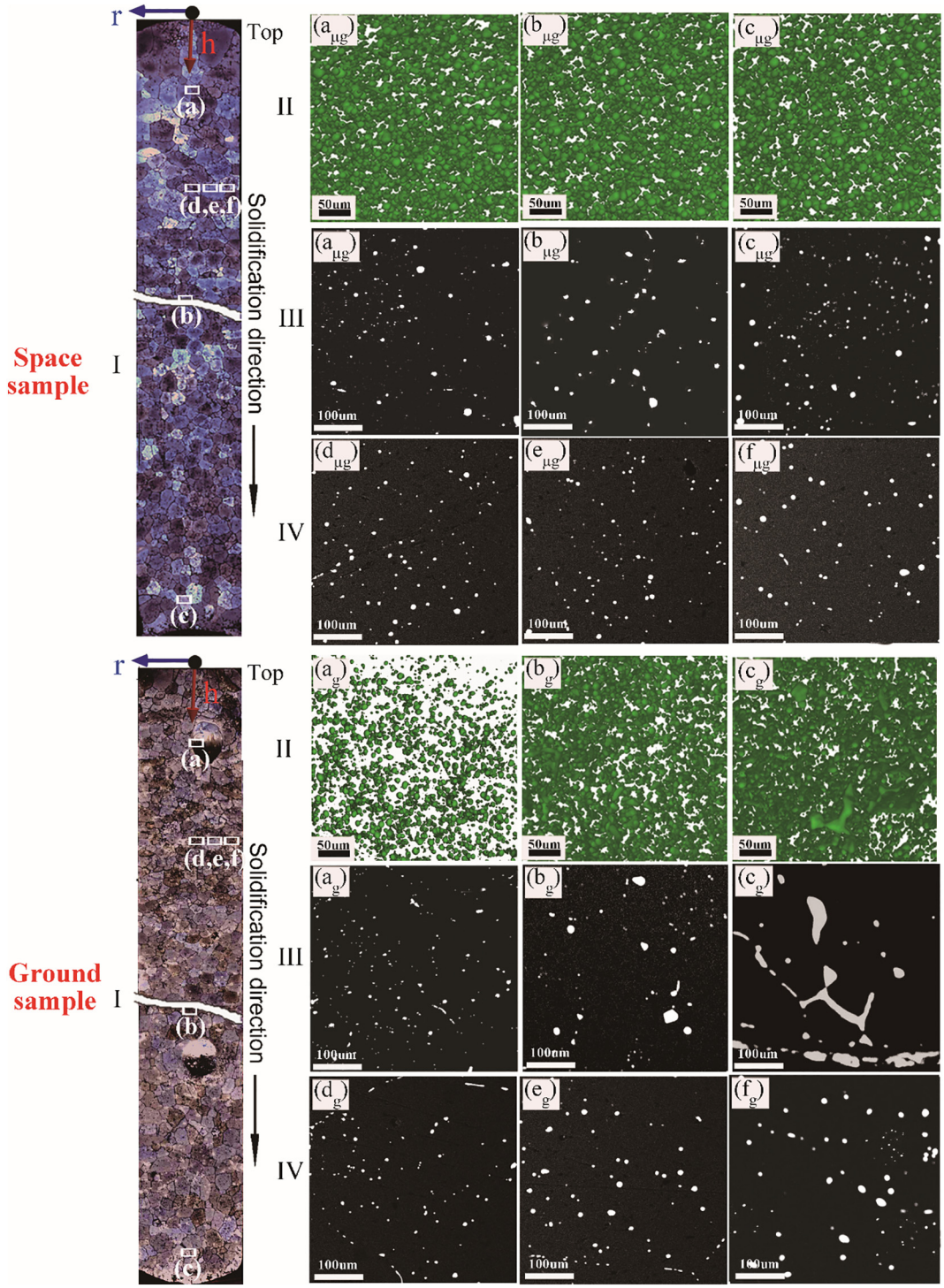


Fig. 1. OM images (I), XRT images (II: a–c), SEM images of the microstructures along the axial direction (III: a–c) and along the radial direction (IV: d–f) of the samples. The corresponding positions are schematically shown in (I). The subscripts g and μg represent respectively the ground and space samples and the same as below.

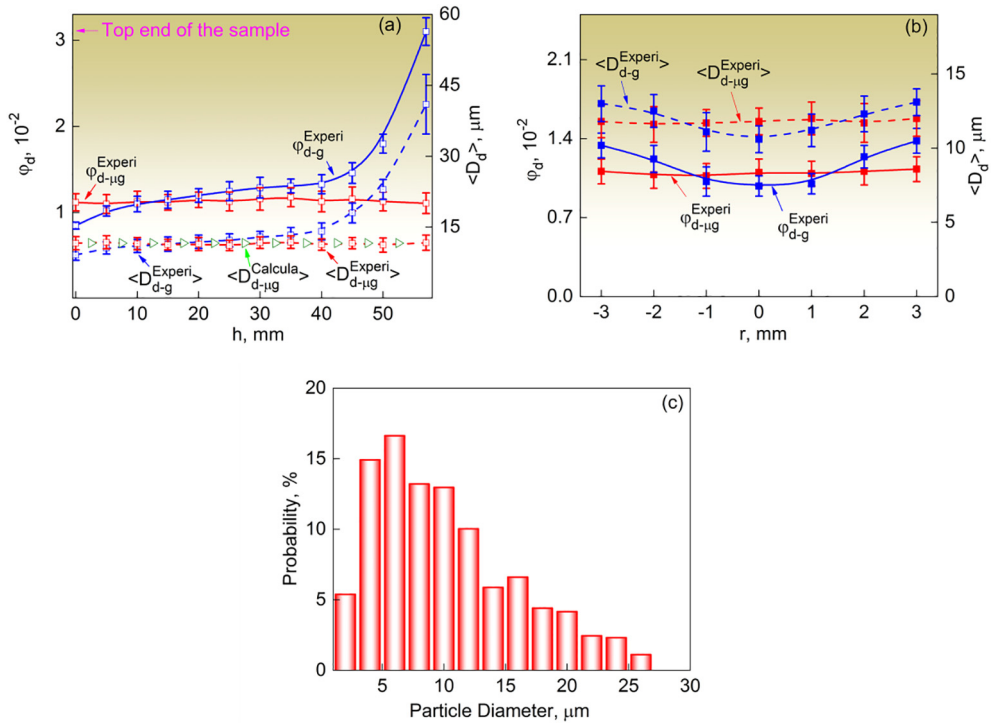


Fig. 2. Volume fraction (φ_d) and average diameter ($\langle D_d \rangle$) of the MPPs along the central axial position h (a) and along the radial direction r at the position of 5 mm from the top end (b) of the samples (see Fig. 1), and size distributions of the MPPs measured for a region of $2000 \times 1500\ \mu m^2$ in the middle part of the space sample (the central coordinate of the region are r = 0 mm, h = 30 mm) (c).

developed by the Chinese Academy of Sciences was used for the quantitative metallographic analysis of the sizes of the MPPs.

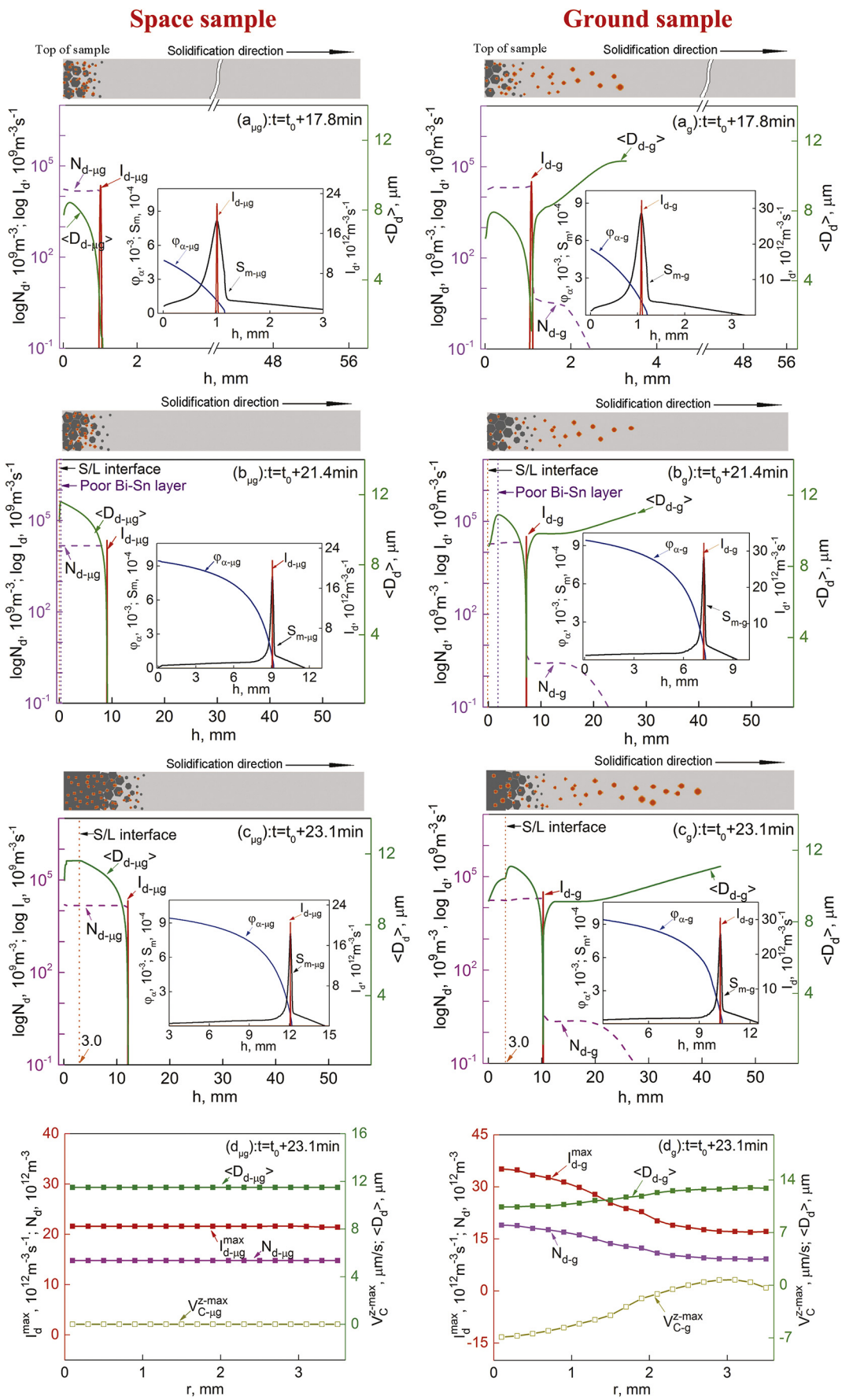
Fig. 1 shows the XRT images as well as the macro- and microstructures of the samples. The EDS analyses demonstrate that the matrix consists of α-Al equiaxed grains and the MPPs are the Bi-Sn-rich phase. The ground sample shows a microstructure with severe phase segregation. The top end region (the first solidified region) is poor in Bi-Sn phase while the bottom end (the last solidified region) is rich in Bi-Sn phase. The morphologies of the MPPs in the top end region of the sample are almost spherical, while some of the MPPs in the bottom end region show an irregular shape. Quantitative metallographic analyses demonstrate that the volume fraction and average diameter of the MPPs distribute in-homogeneously along both the axial direction and the radial direction of the sample, as shown in Fig. 2(a) and (b). In contrast, the sample solidified in space shows a well dispersed distribution of the MPPs throughout the whole sample. Fig. 2(c) shows the typical size distribution of the MPPs in the sample solidified in space. It indicates that the size of the MPPs ranges from 2.0 to 27.0 μm.

A model was developed to describe the solidification microstructure evolution (see Appendix A). The numerical results demonstrate that the temperature gradient along the axial position for the sample solidified on ground is a little larger than that for the sample solidified in space. The average cooling rates from the binodal line temperature (658.9 °C) to the S/L interface temperature (643.5 °C, see Fig. S3 in Appendix A) are ~0.026 °C/s and ~0.029 °C/s for the alloy solidified in space and on ground, respectively.

Fig. 3($a_{\mu g} - d_{\mu g}$) shows the microstructure evolution for the alloy solidifying in space. The melt at the top end of the sample is cooled into the miscibility gap temperature and becomes supersaturated with Bi and Sn first. But no nucleation event of the MPDs occurs there during the whole period cooling through the miscibility gap. When the melt at the top of the sample is cooled into the liquid-liquid-solid (L_1-L_2-S) temperature region, α-Al grains form and the solutes Bi and Sn are expelled into the melt around the α-Al grains. The growth of α-Al causes a quick increase in the supersaturation of the local melt and the nucleation of the MPDs occurs soon (see Fig. 3($a_{\mu g}$)). With the withdrawing of the ampoule, the nucleation region of MPDs advances towards the bottom of the sample. The MPDs move away from the top end of the sample by Marangoni migration and cause the formation of a thin Bi-Sn-poor layer on the top end of the sample (see Fig. 3($b_{\mu g}$)). The S/L interface appears at the top end of the sample at the moment of $t_0 + 21.4$ min. It advances at a rate approximately equal to the withdrawing velocity of the ampoule. All of the MPDs in front of the S/L interface move towards the S/L interface. They are gradually caught by the S/L interface and embed in the matrix. A steady state of solidification soon established after the appearance of the solidification interface. Since that moment, the temperature, concentration and convective flow fields as well as the spatial distribution of the average diameter and number density of the MPDs/MPPs in the melt remain constant until the solidification interface is close to the bottom end of the sample (see Fig. 3 ($c_{\mu g}$)). The numerical result for the average diameter of the MPPs is in a good agreement with the experimental one (see Fig. 2(a)). Calculations demonstrate that the maximum nucleation rate, average diameter and number density of the MPDs/MPPs also distribute uniformly along the radial direction (see Fig. 3($d_{\mu g}$)).

Fig. 3($a_g - d_g$) shows the microstructure evolution for the alloy solidifying on ground. The solidification follows a similar procedure as solidifying in space. That is, no phase transformation occurs during cooling the alloy through the miscibility gap → α-Al grains form and grow during cooling the alloy in the L_1-L_2-S region → the MPDs nucleate, grow

Fig. 3. Calculated nucleation rate (I_{d-i}), number density (N_{d-i}), average diameter ($\langle D_{d-i} \rangle$) of the MPDs/MPPs (a–c), supersaturation of the matrix melt (S_{m-i}), and the volume fraction of α-Al ($\varphi_{\alpha-i}$) (the inserted figures in a–c) along the central axial position h (see Fig. 1) for the alloy solidifying on ground and in space for different time. The maximum axial convective velocity of the melt ($\mathbf{V_{C\text{-}i}^{z-max}}$), the maximum nucleation rate of the MPDs ($I_{d\text{-}i}^{max}$) in front of the S/L interface and the average diameter, the number density in the melt at the S/L interface along the radial direction r of the sample at $t = t_0 + 23.1$ min are shown in (d). Subscript *i* takes μg and g for the sample solidified in space and on ground, respectively.


Space sample
Ground sample
Top of sample
Solidification direction
(a_μg):t=t0+17.8min
(a_g):t=t0+17.8min
(b_μg):t=t0+21.4min
(b_g):t=t0+21.4min
(c_μg):t=t0+23.1min
(c_g):t=t0+23.1min
(d_μg):t=t0+23.1min
(d_g):t=t0+23.1min
S/L interface
Poor Bi-Sn layer
logNd, 10^9m^-3; log Id, 10^9m^-3s^-1
<Dd>, μm
h, mm
r, mm
φα, 10^-3; Sm, 10^-4
Id, 10^12m^-3s^-1
Nd-μg
Id-μg
<Dd-μg>
φα-μg
Sm-μg
Nd-g
Id-g
<Dd-g>
φα-g
Sm-g
3.0
Id^max, 10^12m^-3s^-1; Nd, 10^12m^-3
Vc^z-max, μm/s; <Dd>, μm
Id^max-μg
Vc^z-max-μg
Id^max-g
Vc^z-max-g

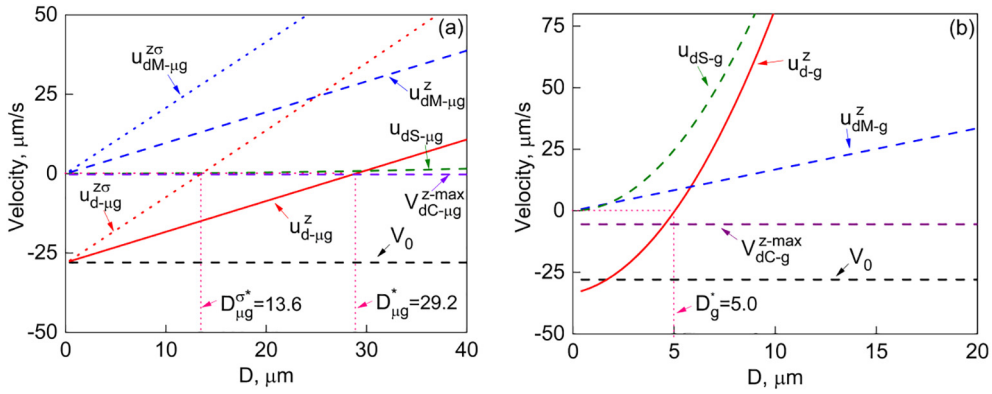


Fig. 4. Z-vector for the resultant ($u_{d\text{-}i}^{z}$), Marangoni ($u_{dM\text{-}i}^{z}$) and Stokes ($u_{dS\text{-}i}^{z}$) moving velocity as a function of the diameter of the MPDs together with maximum convective flow ($V_{C\text{-}i}^{z\text{-}max}$) and the withdraw velocity (V_0) for the alloy solidifying in space (a) and on ground (b) at $t = t_0 + 23.1$ min. The dotted lines are the resultant ($u_d^{z\sigma}$) and Marangoni ($u_{dM}^{z\sigma}$) moving velocity calculated by using the liquid-liquid interfacial energy for Al-Bi alloy. D_i and $D_{\mu g}^{\sigma}$ are the critical diameter of the MPDs determined by $|\mathbf{u_d}^{\mathbf{z}}_{\text{-}i}(D_i)| = 0$ and $|\mathbf{u_d}_{\text{-}\mu g}^{\mathbf{z}\sigma}(D_{\mu g}^{\sigma})| = 0$, respectively. Subscript i takes μg and g for the sample solidified in space and on ground, respectively.

and move → S/L interface appears at the top end of the sample → S/L interface advances towards the bottom end of the sample. What different from the solidifications in space are: 1) the Stokes motion, which is towards the bottom end of the sample, is much stronger. A visible Bi-Sn-poor layer, thus, forms in the top end region of the sample (see Fig. 3 (b_g)); 2) not all the MPDs in front of the S/L interface move towards the S/L interface. Some MPDs of bigger sizes move away from the S/L interface into a relatively high temperature region. The motions of these MPDs make it impossible for the solidification to reach a steady state and lead to the formation of the macro-segregation soon after the beginning of the liquid-liquid phase transformation (see Fig. 3(c_g)). A Bi-Sn-rich region gradually forms in the bottom region of the sample; 3) the convective flow is much stronger (see Fig. S4 in Appendix A). It causes an in-homogeneous distribution of the nucleation rate, average diameter and number density of the MPDs/MPPs along the radial direction of the sample (see Fig. 3(d_g)).

Microgravity conditions may affect the solidification microstructure formation of immiscible alloys through two ways: (1) weakening the convective flow of the matrix liquid; (2) changing the **z** vector of the resultant velocity of the MPDs ($\mathbf{u_d^z}$) by diminishing the Stokes settlement of the MPDs.

Calculations demonstrate that the maximum flow velocity appears at the central axes of the sample (see Fig. S4 in Appendix A). It reaches a value about 6.7 μm/s when solidifying on ground, while it is only about 2.7×10^{-3} μm/s when solidifying in space. The Stokes moving velocity of the MPDs is negligible when the alloy solidifies in space, but it plays the dominant role when the alloy solidifies on ground (see Fig. 4).

Under the directional solidification conditions, an immiscible alloy may form a well dispersed microstructure only if a steady state of solidification establishes. The prerequisite for the establishment of such a steady state of solidification is that the MPDs of all sizes in the melt move towards the S/L interface or $\mathbf{u_d}^{\mathbf{z}}{}_{\forall(R,r,z)} < 0$.

For simplification, we just check the motions of the MPDs located on the central line of the sample. Calculations demonstrate that the MPDs with a diameter smaller 29.2 μm meet the above condition for the alloy solidifying in space ($D_{\mu g}^{*}$ as shown in Fig. 4(a)). This value is larger than the maximum diameter of the MPDs in the melt, which is about the maximum size of the MPPs (~27.0 μm, as shown in Fig. 2c). A steady state of solidification can establish. The sample solidified in space thus shows a homogeneously dispersed microstructure. In order to check the effects of the interfacial energy reduction by the element Sn and the special design of the furnace temperature profile on the microstructure formation, calculations were carried out by using the liquid-liquid interfacial energy for Al-Bi alloy and assuming a common furnace temperature profile ($T_1 = T_2$). The numerical results demonstrate that the critical diameter of the MPDs ($D_{\mu g}^{\sigma *}$) is 13.6 μm. This value is smaller than the maximum diameter of the MPDs in the melt, indicating that a phase segregated microstructure would form. For the alloy solidified on ground, the critical diameter of the MPDs (D_g^{*}) equals 5 μm (see Fig. 4(b)). Both the experimental and numerical results show that this value is much smaller than the maximum diameter of MPDs/MPPs in the alloy solidified on ground, indicating that many MPDs in the melt move away from the S/L interface and a steady state of solidification cannot establish. The sample solidified on ground thus shows a phase segregated microstructure.

The convective flow also affects the distributions of the MPDs/MPPs along the radial direction of the sample. First, the local cooling rate of the melt depends on the convective flow according to Eq. (1):

$$\left.\frac{dT}{dt}\right|_{(z_{nuc},r)} = \left.\frac{dT}{dz}\right|_{(z_{nuc},r)} |\,\mathbf{V_0} + \mathbf{V_C^z}\,| + \left.\frac{dT}{dr}\right|_{(z_{nuc},r)} |\,\mathbf{V_C^r}\,| \tag{1}$$

where $\mathbf{V_0}$ is the withdrawing velocity of the ampoule, $\mathbf{V_C^z}$ and $\mathbf{V_C^r}$ are the **z** vector and **r** vector for the convective velocities. z_{nuc} is the z-coordinate for the position exhibiting the maximum nucleation rate of the MPDs for a given radial position r, T is the temperature and t is time.

Under present directional solidification conditions, the temperature gradient along the radial direction is much smaller compared to that along the axial direction. The cooling rate can be approximated by $\frac{dT}{dt}|_{(z_{nuc},r)} \approx \frac{dT}{dz}|_{(z_{nuc},r)}(\mathbf{V_0} + \mathbf{V_C^z})$. It varies with r in the melt with a convective flow. The nucleation rate of the MPDs thus shows an in-homogeneous distribution along the radial direction of the sample when the alloy solidifies on ground (see Fig. 3(d_g)) and a homogeneous distribution along the radial direction of the sample when the alloy solidifies in space (see Fig. 3($d_{\mu g}$)). Secondly, a motion of the MPDs towards the S/L interface leads to a decrease in their growth/coarsening time, while a motion of the MPDs away from the S/L interface leads to an increase in their growth/coarsening time. The convective flow thus also leads to an in-homogeneous distribution of the MPD diameters along the radial direction of the sample when the alloy solidifies on ground (see Fig. 3(d_g)). Thirdly, the number density of the MPDs depends both on the local nucleation rate and motion of the MPDs. The stronger convective flow when the alloy solidifies on ground also causes an in-homogeneous distribution of the droplet number density along the radial direction of the sample.

In summary, solidifications experiments were performed with Al-Bi-Sn alloys in space and on ground. A model was developed to describe the microstructure evolution during the solidification. The microstructural formation was calculated and discussed. Following conclusions can be drawn: (1) Solidification of immiscible alloys in space may result in the formation of a well dispersed microstructure even if the alloy is solidified at a very low cooling rate if the alloy composition and the solidification schemes are properly designed. (2) Microgravity conditions

can effectively diminish the convective flow of the melt and the Stokes motion of the MPDs and are helpful for the clarification of the kinetic details of the solidification microstructure evolution of immiscible alloys. (3) Convective flow of the matrix melt and the spatial motions of the MPDs are two factors dominating the microstructure formation of immiscible alloys. When solidifying Al-Bi-Sn alloy on ground, the Stokes motion of the MPDs and the convective flow of the matrix melt prevent the establishment of the steady state of solidification and the macro-segregation along the axial direction of the sample occurs soon after the beginning of the solidification. The convective flow of the matrix melt also leads to the macro-segregation along the radial direction of the sample. The weak convective flow of the matrix melt and the diminished Stokes settlement of the MPDs under microgravity conditions are favorable for the Al-Bi-Sn alloy to reach a steady state of solidification and promote the formation of a well-dispersed microstructure.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scriptamat.2018.12.010.

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